Multi-Robot Exploration using Entropy-Based Frontier Selection

Tanvi Vinayak 23349 ; Yashvi Singh 23388

Abstract

This project presents a simulation of multi-robot exploration using an entropy-based frontier selection strategy in a partially known environment. The system models a team of four robots operating on a 100×100 occupancy grid, where each robot incrementally builds its own map through local sensing and shares information with nearby robots.

The exploration strategy is based on selecting frontier regions that maximize expected information gain, while also incorporating reachability constraints to ensure feasible navigation. Robots dynamically assign and update goals, avoid collisions, and coordinate through communication within a fixed range.

Experimental results demonstrate effective coverage of the environment, with progressive map construction and convergence of explored area over time. A key improvement over the baseline approach involves addressing unreachable goal assignments by introducing a reachability-aware penalty during goal selection, which significantly improves exploration efficiency and prevents robot stagnation.

The simulation highlights the effectiveness of decentralized exploration strategies and provides insights into coordination mechanisms in multi-robot systems.

1 Introduction

Autonomous exploration of unknown environments is a fundamental problem in robotics, with applications in search and rescue, surveillance, and planetary exploration. Multi-robot systems offer significant advantages over single-robot systems by enabling parallel exploration, faster coverage, and robustness to individual failures.

A common approach to exploration is frontier-based exploration, where robots identify boundaries between known and unknown regions and move toward them to maximize information gain. Recent approaches enhance this strategy by incorporating entropy-based measures, allowing robots to select goals that maximize expected reduction in uncertainty.

In this project, we implement a multi-robot exploration framework inspired by entropy-based frontier selection. Each robot maintains a local occupancy grid map and explores the environment using local sensing. Robots communicate with nearby agents to share map information, enabling cooperative exploration.

A key challenge addressed in this work is the assignment of reachable goals. The baseline approach may assign goals that are blocked by obstacles, causing robots to become stuck. We address this by incorporating a reachability-aware penalty during goal selection, ensuring that only feasible goals are selected.

2 Problem Setup

The exploration environment is modeled as a two-dimensional occupancy grid of size 100×100. Each cell represents either free space or an obstacle. The global map is used only for simulation and represents the ground truth.

A team of four robots is deployed in the environment. Each robot maintains a local occupancy grid initialized with unknown values (0.5), while free and occupied cells are represented by 0 and 1 respectively.

Each robot is equipped with a circular sensing model with radius 5 cells. Robots can communicate with each other when they are within a communication radius of 10 cells, allowing them to merge their local maps.

The exploration process runs for a maximum of 730 time steps. At each step, robots perform sensing, communication, goal assignment, and motion.

3 Methodology

3.1 Map Representation

The environment is represented using an occupancy grid. Each robot maintains its own local map where unknown cells are initialized to 0.5.

3.2 Sensing

Each robot senses its surroundings within a fixed radius and updates its local map based on observed free and occupied cells.

3.3 Frontier Detection

Frontiers are defined as the boundary between known and unknown regions. These regions are used as candidate goals for exploration.

3.4 Goal Selection

Goals are selected based on an entropy-based measure, prioritizing regions that maximize information gain.

3.5 Communication

Robots share their local maps when within communication range, enabling cooperative exploration.

3.6 Motion and Collision Avoidance

Robots move toward assigned goals while avoiding collisions with other robots.

4 Improvement: Reachability-Aware Goal Selection

A key limitation in the baseline implementation was the assignment of unreachable goals. In some cases, goals were selected behind obstacle regions, making them inaccessible from the robot's current position. This caused robots to become stuck and prevented effective exploration.

To address this issue, a reachability-aware penalty was introduced during goal selection. Goals that are likely to be unreachable are penalized, reducing their likelihood of being selected.

This modification ensures that robots focus on feasible targets and continuously make progress, significantly improving exploration efficiency.

The execution logs show multiple instances of blocked goal detection followed by reassignment, confirming that the reachability-aware modification successfully prevents robots from getting stuck at infeasible targets.

5 System Behavior Analysis

The execution logs provide insight into the internal behavior of the exploration algorithm. At each step, robots identify frontier regions and dynamically assign goals based on the current map.

Frequent goal reassignment is observed as robots continuously update their targets based on newly discovered information. In several instances, candidate goals are identified as blocked, indicating that they are unreachable due to obstacles. These goals are rejected and replaced with feasible alternatives.

Additionally, communication events between robots are observed when they come within communication range. During these events, maps are merged, resulting in a sudden increase in the number of detected frontiers and improved exploration efficiency.

These observations confirm that the system adapts dynamically to the environment and effectively integrates sensing, decision-making, and communication.

5.1 Sample Execution Log

```
robot 2 frontiers found: 3
Robot 2: goal [76 22] blocked
robot 2 reassigning goal → [74 23]

EDGE ADDED (2, 3) → maps merged

robot 3 frontiers found: 39
robot 3 reassigning goal → [76 43]
```

This snippet illustrates: - blocked goal detection - dynamic reassignment - inter-robot communication and map merging

6 Results

6.1 Exploration Progress

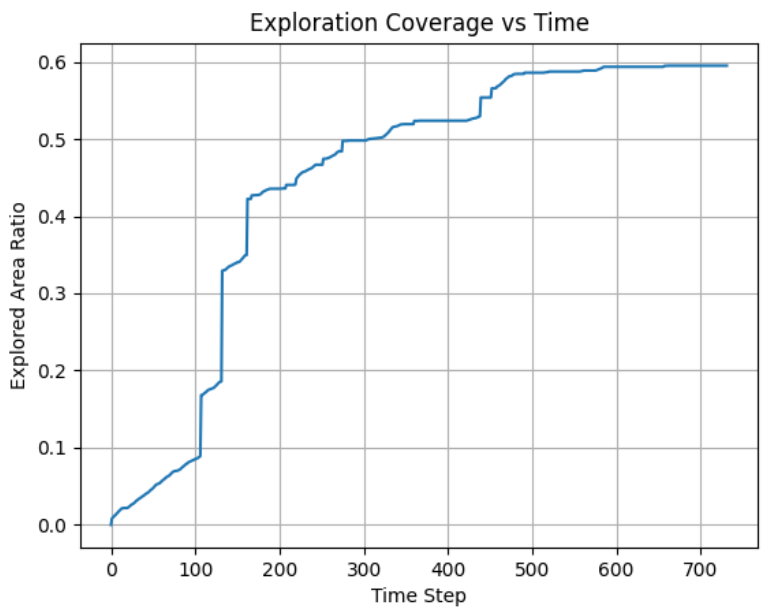
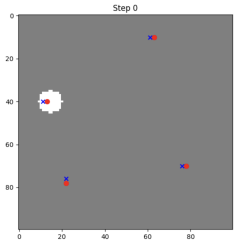


Figure 1: Exploration coverage vs time

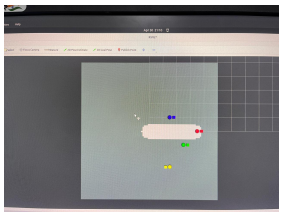
The coverage plot shows a gradual increase in explored area over time. Initially, exploration progresses slowly due to limited sensing. As robots spread out and discover new regions, the coverage increases rapidly. Eventually, the growth slows as fewer unexplored areas remain.

The final coverage reaches approximately 60%, which is expected due to the presence of obstacles and inaccessible regions.

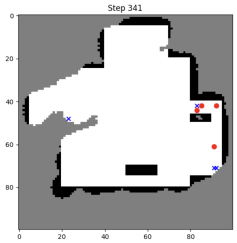
6.2 Exploration Snapshots python vs ROS



(a) Initial

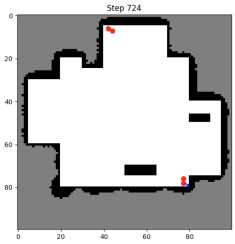


(b) ROS - start of exploration

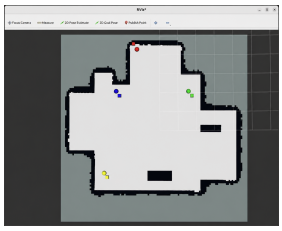


(c) Mid

(d) ROS - during exploration



(e) Final



(f) ROS - final explored

Figure 2: Snapshots of exploration process

The figures show the progressive exploration of the environment. Robots initially explore nearby regions, then spread out and cover larger areas. Communication between robots helps in sharing map information and improving efficiency.

7 Failure Case Analysis

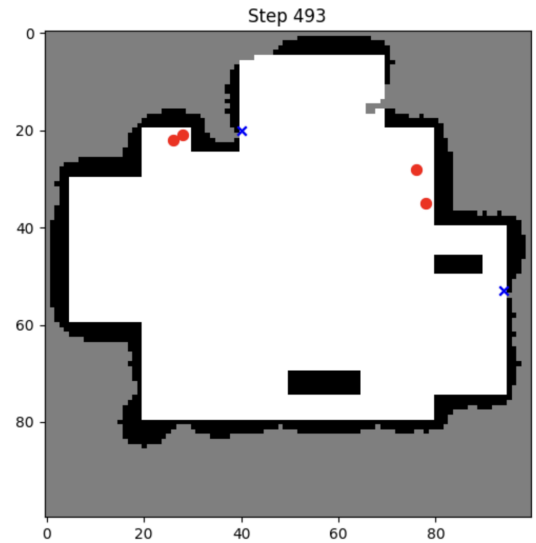


Figure 3: Example of unreachable goal assignment in baseline implementation

In the baseline implementation, robots at approximate coordinates (20,40) were assigned a goal at (20,45) that was unreachable due to obstacles. This resulted in robots getting stuck and failing to explore new regions.

The proposed reachability-aware approach eliminates this issue by rejecting infeasible goals, leading to smoother and more effective exploration.

7.1 Comparison with Baseline Implementation

7.2 Baseline Behavior

In the baseline implementation, goal assignment does not explicitly account for reachability or stability. As a result, robots frequently reassign goals based on nearby frontier regions, which can lead to inefficient exploration.

```
robot 2 frontiers found: 3
robot 2 assigned NEW goal [76 22]
robot 2 reassigning goal → [74 23]
robot 2 reassigning goal → [72 25]
```

The logs show that robots continuously update their goals over time. While this reflects responsiveness to new information, it also indicates instability in the goal selection process, where robots repeatedly switch between nearby targets instead of committing to efficient exploration paths.

Communication between robots is also observed:

```
EDGE ADDED (2, 3) → maps merged
EDGE REMOVED (2, 3)
```

These communication events enable map sharing and contribute to cooperative exploration. However, in the baseline implementation, communication alone does not fully resolve inefficiencies arising from unstable goal selection.

In the later stages of exploration, the logs show:

```
robot 3 frontiers found: 0
robot 3 frontiers found: 0
robot 3 frontiers found: 0
```

This indicates that no unexplored regions remain in the robot's local map, which typically corresponds to the final phase of exploration.

7.2.1 Improved Behavior

In the improved implementation, a reachability-aware penalty is incorporated during goal selection. This allows robots to detect infeasible goals and replace them with alternative feasible targets.

```
robot 2 frontiers found: 3
Robot 2: goal [76 22] blocked
robot 2 reassigning goal → [74 23]
```

Here, unreachable goals are explicitly identified and rejected. As a result, robots are able to avoid ineffective goal assignments and maintain consistent progress during exploration.

7.2.2 Discussion

The comparison highlights the impact of incorporating reachability constraints in multi-robot exploration. While the baseline system demonstrates dynamic goal reassignment, it suffers from instability in goal selection, leading to less efficient exploration.

The improved system addresses this issue by prioritizing feasible goals, reducing unnecessary reassignment, and enabling smoother and more consistent exploration behavior. This improvement is also reflected in the exploration coverage results, which show steady growth without irregular fluctuations.

8 Conclusion

This project demonstrates a multi-robot exploration system based on entropy-driven frontier selection. The system successfully achieves decentralized exploration with communication-based coordination.

The introduction of reachability-aware goal selection significantly improves performance by preventing stagnation. The results show effective coverage of the environment and highlight the importance of combining information gain with feasibility in exploration strategies.